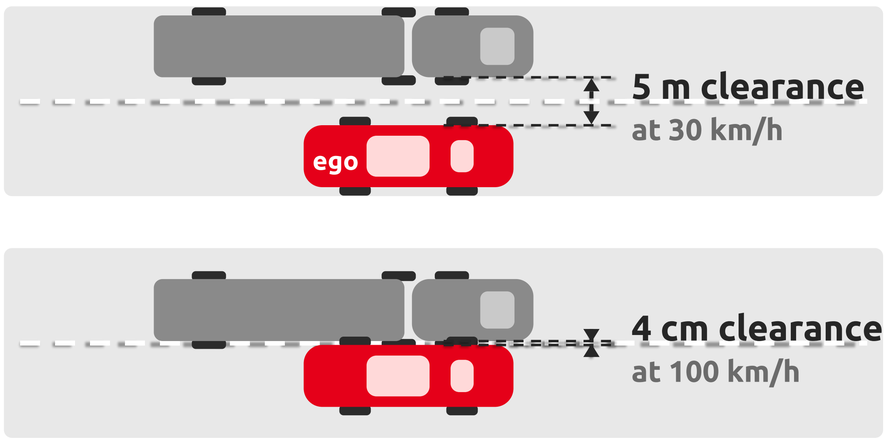
Danger Is a Counterfactual

Toward a Falsifiable Continuous Test Oracle for Autonomous Driving Systems

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Context & Problem

WHAT THE ORACLE SEES

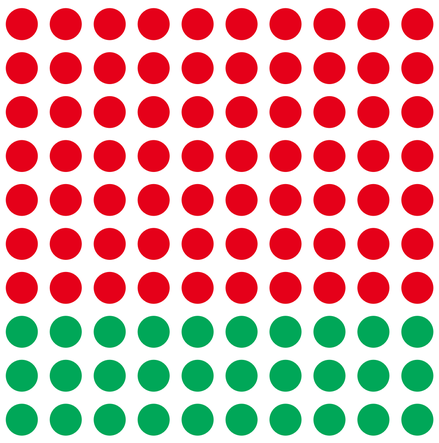


Same verdict for both.
The oracle is one bit: collision / no collision.

WHAT THE NEIGHBOURHOOD SAYS

Re-simulate the 4 cm run 100 times, with the system reacting freshly under small, plausible disturbances:

reaction ± 0.25 s · braking 80–105 % · traffic jitter



70 of 100
end in a 72 km/h head-on

30 no contact

$H(e) = 1/100 \sum \iota(s_i) = 0.53$
severity-weighted share of worlds that crash

one dot = one plausible neighbouring world

A near-miss is recorded as a PASS, while 70 % of its neighbouring worlds are a violent crash.
That single "passing" run carries **a third of the entire 150-run suite's harm.**

WHY THE OBVIOUS FIX FAILS

Report a continuous danger score instead? **Two obstacles.**

1 Proximity is not danger

TTC RANKING	TRUE HARM · 100 REPLAYS
#1 Run A TTC 0.45 s, oncoming car	#1 Run B rear-ends in 10 of 100
#2 Run B TTC 2.3 s, "5× safer"	#2 Run A 0 crashes in 100

ORDER REVERSED

A run TTC calls the most critical in the suite never crashes in 100 replays: the ego was in control. A run it rates 5× safer rear-ends in one replay in ten.

TTC describes the world that **happened**, not the ones next door. It cannot tell control from luck.

2 Nothing to validate against

the oracle says **danger = 0.7** — *check it against what?* → what the run shows: **no injury**

A verdict no observation can contradict is a convention, not a measurement.

This is why the field still runs on one bit: every continuous score we are aware of is unvalidated on the very run it judges.

MEASURE IT BY REPLAYING IT

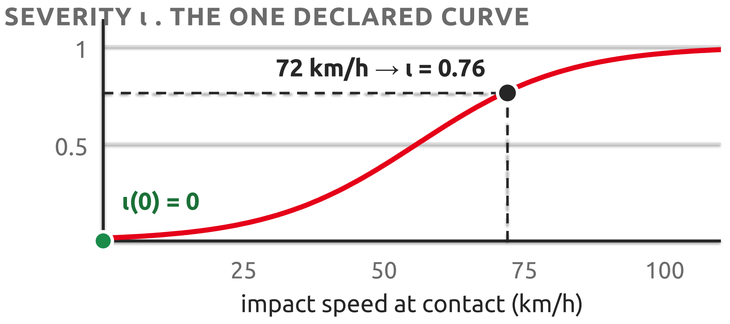
Why do we all call the 4 cm miss dangerous when nothing happened? Because it **almost** happened, and in simulation, those neighbouring worlds are computable.

$$H(e) = \frac{1}{N} \sum \iota(s_i)$$

N replays, drawn from a declared kernel κ — **ι: impact speed → injury weight, with ι(0) = 0**

The harm of a run is the severity-weighted fraction of its plausible neighbourhood that ends in a crash.

SEVERITY ι . THE ONE DECLARED CURVE



Falsifiable. "This run's harm is 0.53" is confirmed or refuted by more replays. No score of §1 can be wrong.

Subsumes the verdict. Shrink the disturbances to zero, make ι a step, and H is exactly the collision bit.

The kernel is declared. Harm is reported kernel-indexed and auditable, like the controllability parameter of ISO 26262-3.

THE PRACTICAL ORACLE

Replays cost simulation time. So **measure once, then predict:** the per-run tier needs no simulation beyond the run testing already executes.

REFERENCE TIER . MEASURE

150 scenarios × 100 replays each
↓
15,000 simulations → true harm H — ≈ 800 GPU-h, once

↓ *distil*

SINGLE-RUN TIER . PREDICT

the recorded run: no extra simulation
↓
gradient-boosted trees (15,000 replay outcomes) · forest on log-harm (scenario-level labels) · zero-parameter physics (flags rare catastrophes)
↓ *rank fusion*
predicted H + 90 % conformal interval

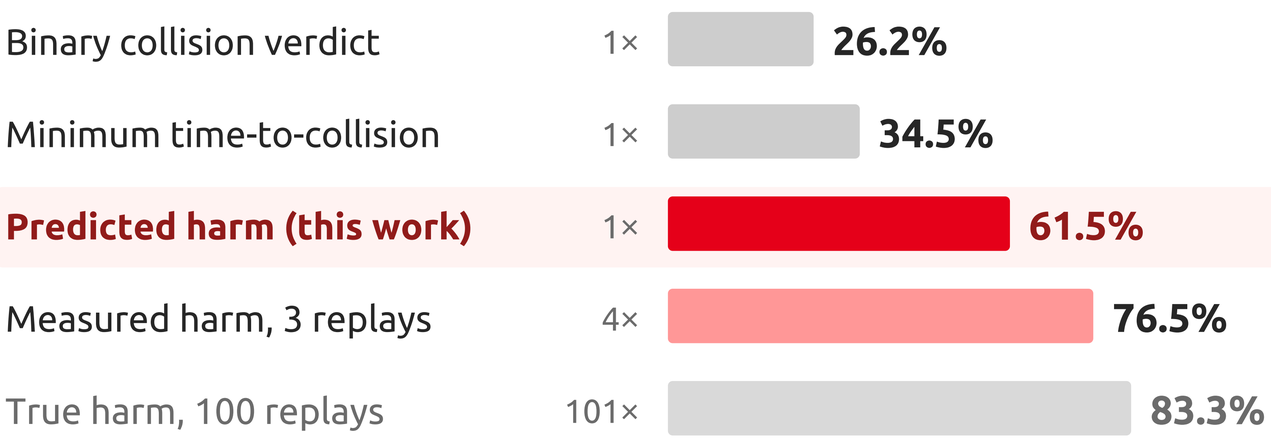
Ground truth is computed, not elicited, so validation needs no human judgment. Subject: openpilot in MetaDrive, 150 scenarios from 5 NHTSA pre-crash templates.

800 GPU-hours, once. Distilled, scoring a new run takes milliseconds; replays are reserved for confirming what the estimate escalates.

What it does not claim. H is defined relative to a declared κ. A different kernel is a different, equally auditable question.

DOES IT WORK?

SHARE OF THE SUITE'S HARM IN THE 15 RUNS (OF 150) YOU INSPECT



Model rows out-of-fold, splits at scenario level. Cost is simulation cost per run.

$r_s = 0.75$
Ranks runs as well as three-replay measurement, at a quarter of its simulation cost.

94 %
of its nominal 90 % conformal intervals contain the re-simulated truth: stated confidence kept, erring safe.

0.97 – 1.00
power on planted severity regressions, where the binary verdict stays at chance (0.03 – 0.09) at every budget.

TAKEAWAY

A test oracle for a stochastic safety-critical system should report not whether the observed world went wrong, but **how much of the adjacent worlds do**.

FUTURE WORK

Replication in CARLA on a second driving stack. Does the estimate transfer, or is it learning simulator artefacts?

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